DF2Net: A Differential-Feature Dynamic-Fusion Network for Building Change Detection in Remote Sensing Images

Like Zhao, Mingwang Yang, Yiran Zhao, Yuqing Liu, Huawei Jiang, and Zhen Yang

***Abstract*—With the continuous advancement of urbanization, the importance of building change detection in urban planning, disaster management, and other fields has become increasingly prominent. However, existing change detection methods still have significant limitations in spatial domain difference feature representation and multiscale difference capture. To address this, we propose a novel differential-feature dynamic-fusion network (DF2Net), which aims to improve the accuracy of building change detection in remote sensing images. DF2Net achieves efficient detection of change areas through the collaborative design of three key modules: differential-feature frequency-domain attention module (DFAM), global–local feature dynamic-fusion module (GFDM), and pseudo-label deep-supervision module (PDM). Specifically, DFAM combines the Fourier transform and self-attention mechanisms to extract key information from the frequency domain, thereby improving the feature representation of change areas. GFDM dynamically fuses global and local features to effectively capture multiscale change information. PDM optimizes feature learning in the middle layer of the decoder by introducing pseudo-label supervision. Experimental results on three mainstream building change detection datasets, LEVIR-CD, WHU-CD, and HRCUS-CD, show that DF2Net significantly outperforms existing mainstream methods in both *F*1-score and IoU, demonstrating superior performance. This indicates that DF2Net has broad application prospects in handling complex scenarios and multiscale change detection tasks. The code available from https://github.com/Mw-yang/DF2Net.**

***Index Terms*—Change detection, deep learning, differential features, fast Fourier transform (FFT), pseudo label, remote sensing images.**

I. Introduction

CHANGE detection plays a crucial role in remote sensing image processing, focusing on extracting differential information between multitemporal or dual-time images of the same area. It finds broad applications in domains like land cover monitoring [1], disaster management [2], and environmental monitoring [3]. As urbanization continues to advance, there has been an increasing emphasis on building change detection, particularly in areas like identifying illegal constructions [4] and urban expansion [5]. In recent years, the rapid development of remote sensing technology has made it easier to obtain high-resolution remote sensing images [6]. However, due to the complexity of building changes, the diversity of scenes, and the interference of environmental factors (such as seasons, weather, lighting, etc.), accurately and efficiently detecting building change information remains a significant challenge [7], [8].

Over the last few decades, a wide range of studies have been carried out by researchers focusing on methods for change detection. These methods can be classified into two broad categories based on how features are extracted: traditional approaches and deep learning techniques. Traditional building change detection methods can be further divided into three types: methods based on image algebra [9], [10], methods based on image transformation [11], [12], and postclassification comparison methods [13], [14]. Methods based on image algebra usually rely on algebraic operations to classify images at the pixel level, while methods based on image transformations highlight change information by performing statistics and transformations on images. Both types of methods are direct classification methods, which are simple and easy to implement, but with the increasing complexity of semantic information in remote sensing images, they have significant shortcomings in dealing with change information in complex scenes. Whereas the postclassification comparison method refers to extracting building features from images captured at different time points and comparing these extracted features to detect changes. While the postclassification comparison method yields improved detection outcomes, it is cumbersome and highly reliant on the accuracy of the building extraction process.

In recent years, with the rapid development of deep learning, convolutional neural networks (CNNs) have provided a novel solution for building change detection [15]. For instance, Liu et al. [16] introduced the local-global pyramid network (LGPNet), which utilizes a feature pyramid module to capture building objects across multiple scales and perspectives, effectively preventing the omission of buildings with diverse sizes and shapes. Zhang et al. [17] developed the deep supervision image fusion network (DSIFN), which extracts deep features from dual-temporal images using the VGG network

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The authors are with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou 450001, China (e-mail: like_zhao@haut.edu.cn; ymw@stu.haut.edu.cn; yiran_zhao2022@163.com; liuyuqinglucky@163.com; lhwcad@126.com; 593797596@qq.com).

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and further refines them with spatial and channel attention modules, enhancing the performance of change detection. Chen et al. [18] proposed the dual attention Siamese network (DASNet), which employs attention mechanisms to capture long-range dependencies between pixels, thereby effectively extracting change features. Shi et al. [19] developed the deep supervision attention metric-based network (DSAMNet), which combines a deep supervision module and a metric module to learn change features. The model incorporates the convolutional block attention module (CBAM) to suppress irrelevant features, enhancing the accuracy of the results. Jiang et al. [20] introduced MDANet, which features a difference module for learning initial change information and an attention refinement module that optimizes these difference features, enabling more effective extraction of image change information. However, the localized nature of convolution operations restricts these methods' capacity to capture global interactions between distant pixels.

Since Transformer was proposed, it has seen remarkable success in natural language processing (NLP) [21] and computer vision (CV) [22], owing to its superior ability to represent global features. In recent years, it has been increasingly applied in remote sensing image processing, yielding impressive results [23]. For instance, Chen et al. [24] proposed the BiT, the first application of Transformer technology in the field of change detection. This method utilizes the self-attention mechanism to capture long-range contextual dependencies in dual-temporal images, surpassing CNN-based methods in performance. Bandara and Patel [25] proposed the Transformer-based ChangeFormer model, which employs a hierarchical Transformer encoder to extract features at multiple layers, computes differences, and passes them through an MLP decoder to produce change predictions. Likewise, Zhang et al. [26] developed the SwinSUNet, a pure Transformer model built upon the Swin Transformer, showcasing considerable promise in change detection tasks. Additionally, Feng et al. [27] proposed a dual-branch framework that integrates CNN and Transformer for feature extraction. It incorporates intrascale interaction and interscale feature fusion modules to effectively integrate features from both branches. Additionally, Zhang et al. [28] introduced the deep multiscale multiattention Siamese transformer network (DMMSTNet), which integrates contextual attention within the Siamese encoder and leverages Transformer to improve global feature interactions. Song et al. [29] proposed the CCLNet, which uses a lightweight backbone for semantic feature extraction and incorporates a temporal context-aware Transformer to improve feature correlation across different images. In contrast to Siamese-based architectures, Xue et al. [30] proposed a three-branch model designed to capture long-range relationships of change objects across various scales. By adding the full-scale Swin-Transformer as the third branch with dense connections, the network significantly enhances the accuracy of change detection. Although most deep learning methods perform well in spatial domain feature extraction, these methods usually ignore the unique advantages of frequency domain features in suppressing noise and capturing subtle changes. Recently, some related studies have begun to explore the combination of frequency-domain features and deep learning. For example, Zheng et al. [31] proposed a high-frequency attentional mechanism that improves the ability to localize buildings by enhancing the high-frequency information and improving the accuracy of change detection. Liang et al. [32] utilized the self-attention mechanism to construct an attention module containing both high-frequency and low-frequency branches, respectively, and subsequently fused the high-frequency and low-frequency features to enhance the representation of the change features. Liu et al. [33] designed the multi-neighborhood frequency gate attention, which utilizes the frequency-domain analysis to effectively reduce the interference caused by the difference in image styles, improves the degree of attention to the target's location information, and demonstrates a certain advantage in the change detection task.

In summary, thanks to the powerful feature extraction capabilities of deep learning, existing change detection methods have made significant progress, but still face some key challenges. First, the synergy between frequency-domain attention and differential features is not fully explored, which limits the change detection ability in complex scenes. In addition, when dealing with multiscale change scenarios, it is difficult for most methods to effectively integrate global and local features, resulting in insufficient utilization of the relationship between global and local features, thus affecting the capture of multiscale change information. Meanwhile, existing models often depend on a single final supervision, causing a significant discrepancy between features at intermediate layers and the output. This discrepancy adversely affects both model convergence speed and the efficiency of parameter updates in intermediate layers. To address the above problems, we propose a novel differential-feature dynamic-fusion network (DF2Net). The main contributions of this article are as follows.

1) We propose the differential-feature frequency-domain attention module (DFAM) for capturing change information, which combines the Fourier transform with a self-attention mechanism to extract critical information from the frequency domain, thereby enhancing the feature representation of change regions.
2) The global–local feature dynamic-fusion module (GFDM) is designed to effectively aggregate global and local features through the dynamic fusion mechanism, thereby capturing richer multiscale change information.
3) We introduce the pseudolabel deep-supervision module (PDM), which utilizes pseudolabels to supervise intermediate layers of the decoder. This reduces the feature discrepancy between intermediate layers and the final output, thereby enhancing both the accuracy of change detection and the model's convergence speed.

The rest of the article is organized as follows. Section II introduces the details of the proposed method. Section III presents a series of experimental results and a detailed analysis. Section IV contains a discussion of the experimental results. In Section V, we summarize the work presented in this article.

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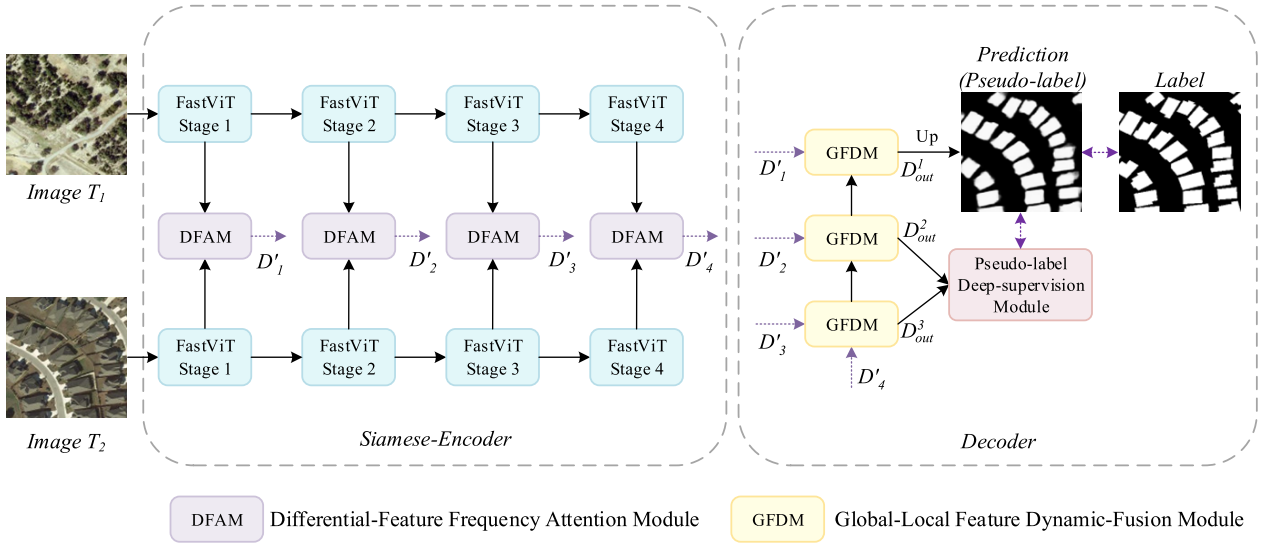


Fig. 1. Architecture of the proposed DF2Net.

II. Method

This section begins with an introduction to the proposed structure of DF2Net. Then, the DFAM module, the GFDM module, and the PDM module are described in detail.

A. Architecture Overview

As illustrated in Fig. 1, the proposed DF2Net is an end-to-end network with a Siamese encoder, where the encoder employs FastViT [34] for feature extraction. Specifically, the bitemporal remote sensing images are input into the dual branches of the Siamese encoder to generate the corresponding four-stage feature maps. Subsequently, the dual-branch feature maps at each stage are input into the DFAM, which extracts the initial difference features from the bitemporal images and further refines these features. In the decoder part, the GFDM module dynamically fuses the multilevel difference features and progressively restores them to the original resolution layer by layer. After processing through three GFDM modules, the fused features D^1_{out} are upsampled to generate the final prediction results, and the prediction loss L_{pre} is calculated with the label. Meanwhile, to improve the convergence speed and learning capacity of the model, the outputs D^2_{out} and D^3_{out} of the first two GFDM modules are input into the PDM module to obtain the middle prediction results. The middle predictions are then compared to the model's final predictions (pseudolabels) to calculate the pseudolabel supervision loss L_{pse}. This process enables the model to progressively enhance its predictive capabilities throughout the iterations, providing further guidance for subsequent learning steps. Overall, DF2Net effectively captures the differential information between bitemporal remote sensing images and enhances change detection accuracy through the dynamic fusion mechanism of multilevel features.

B. Differential-Feature Frequency Attention Module

In change detection tasks, accurately capturing the differential features of bitemporal remote sensing images is essential for precise predictions. However, due to factors such as illumination and seasonal variations, significant spectral differences often exist between bitemporal images, complicating the extraction of differential features. Existing methods primarily focus on feature extraction in the spatial domain, potentially neglecting important variations in the frequency domain. To overcome this challenge, this article proposes the DFAM module. By transforming the differential features into the frequency domain and incorporating a self-attention mechanism, DFAM learns the global interaction information of pixels in the frequency domain, thereby improving the detection of detailed changes.

The structure of the DFAM is shown in Fig. 2. Specifically, the feature maps T_1 and T_2 from the bitemporal images are subtracted, and the absolute value is computed to obtain the initial differential feature D_i

$$D_i = |T_1 - T_2|. \tag{1}$$

Three 1×1 convolutions are applied to the difference feature D_i to obtain three feature tensors: Query ($Q \in \mathbb{R}^{C\times H\times W}$), Key ($K \in \mathbb{R}^{C\times H\times W}$), and Value ($V \in \mathbb{R}^{C\times H\times W}$). These features are used for subsequent attention calculations. Next, the fast Fourier transform (FFT) is applied to the two features Q and K, converting the spatial domain features into frequency domain representations

$$\begin{cases} Q' = \text{FFT}(Q) \\ K' = \text{FFT}(K). \end{cases} \tag{2}$$

In the frequency domain, the similarity between features Q' and K' is calculated using dot product operations to obtain the initial frequency-domain attention. The frequency-domain attention is then transformed back to the original spatial domain using the inverse FFT (IFFT).

The Softmax function is then applied to model the long-distance pixel dependencies of the features, yielding the attention Att′

$$\text{Att}' = \text{Softmax}(\text{IFFT}(Q' \odot K')). \tag{3}$$

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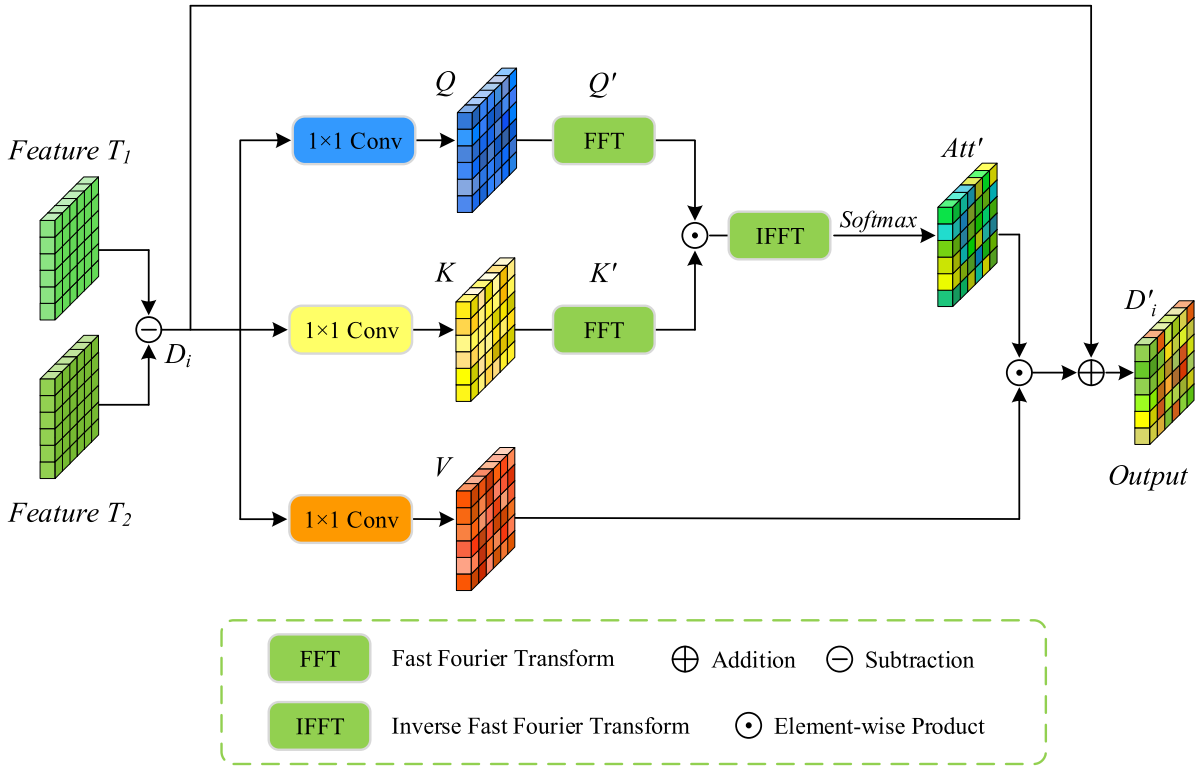


Fig. 2. Differential feature frequency attention module.

The feature V is weighted using the attention Att'. In addition, to enhance feature reuse and avoid the phenomenon of gradient vanishing, the residual connection is utilized to sum the initial differential feature D_i and the spatial-domain feature pixel by pixel to obtain the enhanced differential feature D'_i

$$D'_i = D_i + \mathrm{Att}' \odot V. \tag{4}$$

By learning the global interaction information of the bitemporal image in the frequency domain, the DFAM module enhances the perception of differential features, allowing the model to capture subtle differences more accurately and improving change detection.

C. Global–Local Feature Dynamic-Fusion Module

Bitemporal remote sensing images typically exhibit spatial variations at multiple scales, so it is difficult to rely on a single level of feature representation (global or local) to simultaneously take into account the detailed and overall variation information. Although global and local features complement each other well, considering practical issues such as image alignment bias, the simple fusion of bitemporal features is not only difficult to utilize the complementary advantages of the features, but also may bring negative impacts. To address this problem, we propose the GFDM module, which separately extracts features from the global and local levels, and adaptively aggregates them using the dynamic fusion mechanism. This process enhances the complementarity between features, enabling the capture of richer and accurate multiscale change information.

As shown in Fig. 3, two different levels of difference features D'_i and D'_{i-1} are input into the global feature and local feature unit, respectively. The Global Feature Unit captures the extensive contextual information through the Swin-Transformer Block [35] to generate the global feature F_G, while the local feature unit utilizes a convolution operation to extract the local detail change information to generate the local feature F_L.

To achieve dynamic fusion of global and local features, we introduce the Sigmoid function to calculate the pixel-level weight of local features W_L. Specifically, local feature F_L generates weights through the Sigmoid function

$$W_L = \mathrm{Sigmoid}\left(F_L\right). \tag{5}$$

The weight W_G of the global features is calculated by $1-W_L$

$$W_G = 1 - W_L. \tag{6}$$

This design is inspired by SKNet [36], assigning adaptive weights to different features to maximize their complementary effects, thereby dynamically adjusting the relative importance of global and local features during feature fusion. The global feature F_G and local feature F_L are dynamically weighted by their corresponding weights W_G and W_L, respectively, and then the weighted features are concatenated

$$F_{\mathrm{weighted}} = \mathrm{Cat}\left(W_G \otimes F_G, W_L \otimes F_L\right) \tag{7}$$

where Cat is the concatenation operation along the channel dimension.

The concatenated features are further refined by 3×3 DWConv to generate the dynamic fusion feature F_D

$$F_D = f^{3\times 3}\left(F_{\mathrm{weighted}}\right). \tag{8}$$

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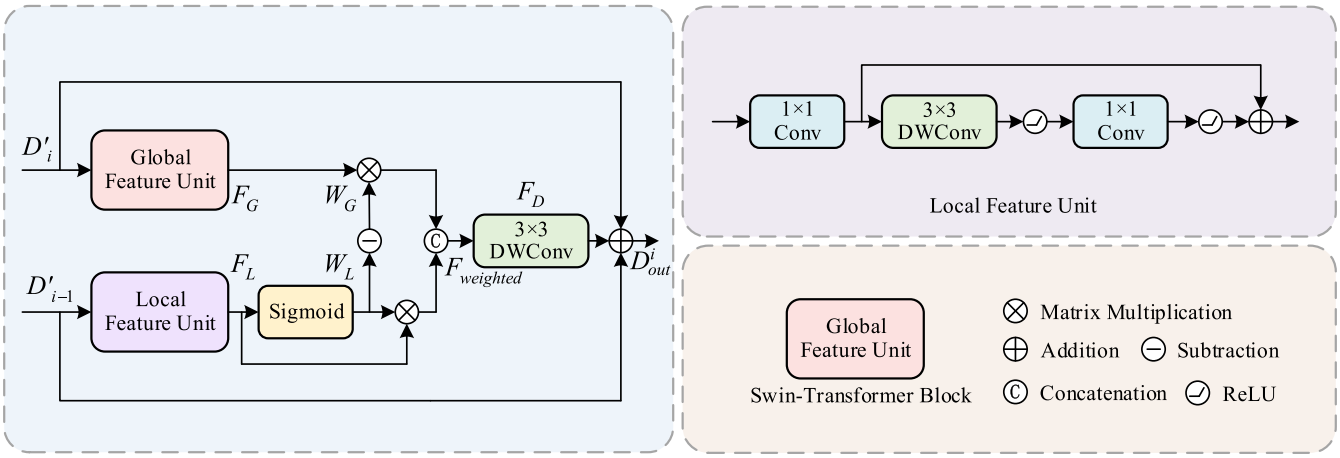


Fig. 3. GFDM.

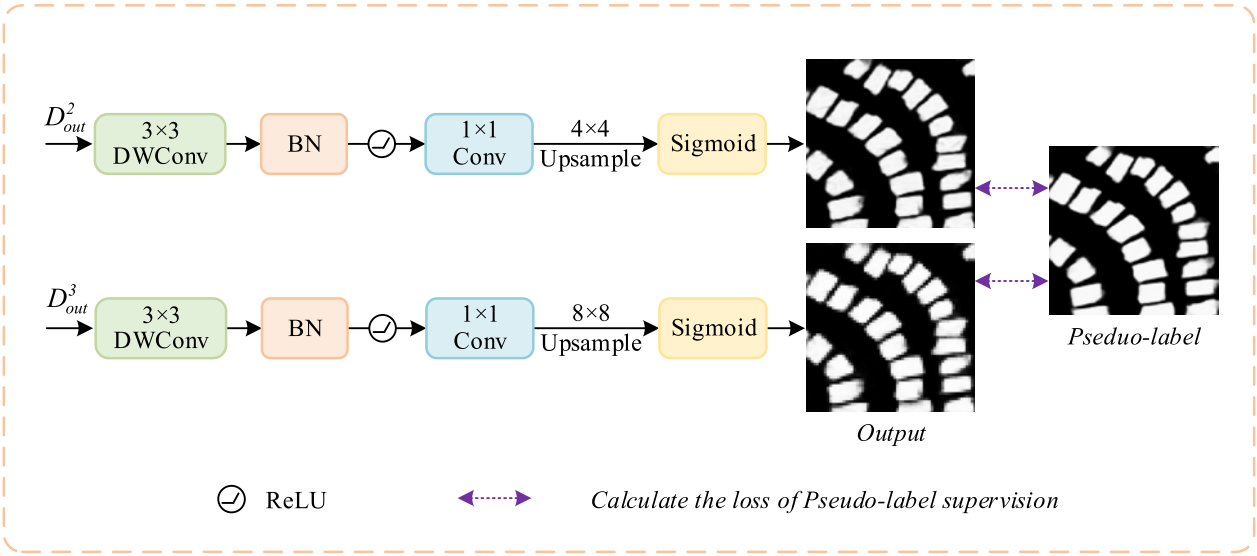


Fig. 4. PDM.

Finally, the dynamic fusion feature F_D is fused with the initial input features D'_i and D'_{i-1} through a residual connection to obtain the module's final output

$$D^i_{\text{out}} = F_D + D'_i + D'_{i-1}. \tag{9}$$

Through this dynamic weight allocation and fusion mechanism, the GFDM module can adaptively adjust the relative importance of global and local features, thereby enhancing the complementarity between features and effectively capturing multiscale change information.

D. Pseudolabel Deep-Supervision Module

Existing deep learning models typically supervise only the final output layer during training. For more complex networks, this approach can result in issues like vanishing gradients during backpropagation, affecting the stability of the middle layer parameters and weakening the model's capability to detect change regions. To address this issue, this article proposes the PDM module, designed to provide additional supervision for the middle layers, enabling the model to learn effective features more quickly, thereby accelerating convergence and enhancing overall performance.

As shown in Fig. 4, features D^2_{out} and D^3_{out} are input into the PDM module, where operations such as DWConv, BN, and ReLU are used to extract features. The extracted features are subsequently upsampled to match the size of the final output, and the Sigmoid function is applied to generate the corresponding prediction results.

P_1 and P_2 denote the output features of the two layers, respectively, $f^{n\times n}$ is a $n \times n$ convolution, Up indicates the upsample operation, and the calculation process can be expressed as follows:

$$P_i = \text{Sigmoid}\left(\text{Up}\left(f^{1\times 1}\left(\text{ReLU}\left(\text{BN}\left(f^{3\times 3}\left(D^i_{\text{out}}\right)\right)\right)\right)\right)\right). \tag{10}$$

The two prediction results are calculated using the loss function with the pseudo labels, respectively, and participate in the back-propagation process, thereby helping the model effectively constrain the middle layer features during the training process.

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Unlike common deep supervision strategies, the PDM module leverages the network's final output as pseudolabels to guide feature learning in the middle layers. This method essentially establishes a "target orientation" for the model's middle layers, ensuring that feature extraction in these layers aligns better with the final output, thereby reducing feature bias. Particularly in complex scenarios, using the final output as pseudolabel can explicitly guide the middle layers on which features are most relevant, helping the model adjust the parameters of these layers to progressively approach optimal feature representations.

By employing this pseudolabel supervision mechanism, the PDM module not only improves the accuracy of prediction results but also facilitates the progressive refinement of feature representations, thereby improving the model's capacity to discriminate change regions.

In this article, we use the binary cross-entropy loss (BCE Loss) function to calculate the loss. The computation of BCE Loss is shown in (7)

$$\text{BCE Loss}(p, y) = -\sum_{i}^{N} \left(y_i \log(p_i) + (1 - y_i) \log(1 - p_i)\right) \tag{11}$$

where N is the number of pixels, y_i and p_i denote the label and prediction result of the ith pixel, respectively.

The loss function in this article consists of two parts: the prediction loss L_{pre} and the pseudolabel supervision loss L_{pse}. Therefore, the total loss function can be expressed as follows:

$$L_{\text{total}} = \alpha L_{\text{pre}} + \beta \sum_{i=1}^{2} L_{\text{pse}}^{i} \tag{12}$$

where α and β balance the two losses, and their values will affect model performance. The optimal values will be determined in the following discussion.

III. Experiment and Analysis

A. Dataset Description

To verify the effectiveness of the proposed method, LEVIR-CD [37], WHU Building Change Detection Dataset [38] (WHU-CD), and HRCUS-CD [39] were selected for experimental analysis.

1) LEVIR-CD: This dataset includes 637 pairs of bitemporal images, each with a spatial resolution of 0.5 m and a pixel size of 1024 × 1024. It covers a variety of building types, including villas, high-rise apartments, small garages, large warehouses, and more, and can effectively reflect the changes in complex buildings. Due to the GPU memory limitations, the original images were cropped into 256 × 256 patches. In the end, there were 7120 training samples, 2048 test samples, and 1024 validation samples.

2) WHU-CD: This dataset contains a pair of aerial images with a pixel size of 32 507 × 15 354, captured in 2012 and 2016. The images cover regions in New Zealand where building changes occurred due to urban reconstruction after earthquakes. Similarly, it was cropped to a size of 256 × 256 pixels to accommodate the experimental equipment, resulting in 6096 training samples, 762 test samples, and 762 validation samples.

3) HRCUS-CD: The dataset consists of 11 388 pairs of 256 × 256 images, each with a resolution of 0.5 m. All images were collected in Zhuhai, China, covering a wide range of complex changes such as villages, industrial parks, farmland, and high-rise buildings in the city, thus enhancing the diversity and representativeness of the dataset. After division, there were 7974 training samples, 1138 test samples, and 2276 validation samples.

B. Experimental Details

All experiments in this article are conducted on NVIDIA GeForce RTX 4090 24 GB. Both DF2Net and the comparison methods are constructed using the Pytorch framework. Data augmentation techniques applied during preprocessing included random hue saturation values, random scaling, random horizontal-vertical flipping, and random rotation by 90°. Then crop the image to 256 × 256 pixels and input it into the network with a batch size parameter of 8. During training, the Adam algorithm [40] is utilized to optimize the network. The initial learning rate of DF2Net is set to 0.0005, and the learning rate is set using the cosine annealing algorithm [41] with a period of 10 epochs. DF2Net is trained for 100 epochs on three datasets: LEVIR-CD, WHU-CD, and HRCUS-CD. The hyperparameters of the compared methods in this article follow the settings of the original paper.

C. Evaluation Metrics

In this article, we select five evaluation metrics, overall accuracy (OA), precision, recall, $F1$, and intersection over union (IoU), to evaluate the performance of DF2Net and other comparison networks. The formulas for these evaluation metrics are shown in the following equations:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{13}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{15}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{16}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}. \tag{17}$$

D. Comparison of State-of-the-Art Methods

In this section, we conduct a series of comparison experiments to verify the effectiveness of the proposed DF2Net. Specifically, we compare DF2Net with existing methods, including FC-Siam-conc [42], SNUNet [43], ICIF-Net [27], LSNet [44], AERNet [39], AMTNet [45], GeSANet [46], TinyCD [47], CDNeXt [48], SEIFNet [49], ISDANet [50], and EFICNN [51]. All networks are experimented on three datasets. In our experiments, the best results for each metric are highlighted in bold, and the second-best results are underlined.

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TABLE I
COMPARISON WITH OTHER METHODS ON THE LEVIR-CD DATASET

Methods	Publication	OA(%)	P(%)	R(%)	F1(%)	IoU(%)
FC-Siam-diff	2018, ICIP	96.33	84.93	81.61	83.23	71.28
SNUNet	2022, GRSL	97.77	90.97	88.81	89.88	81.62
ICIF-Net	2022, TGRS	97.57	89.61	88.47	89.04	80.24
LSNet	2022, IGARSS	97.76	89.73	90.21	89.97	81.77
AERNet	2023, TGRS	97.92	91.30	89.98	90.63	82.87
AMTNet	2023, ISPRS	97.64	89.88	88.89	89.38	80.81
GeSANet	2023, TGRS	97.85	91.08	89.53	90.30	82.31
TinyCD	2023, NCA	97.62	90.24	88.17	89.19	80.49
CDNeXt	2024, JAG	97.11	87.59	86.30	86.94	76.89
SEIFNet	2024, TGRS	97.97	90.94	90.87	90.91	83.33
ISDANet	2025, TGRS	98.01	91.19	90.91	91.05	83.58
EFICNN	2025, TGRS	98.03	91.80	90.51	91.15	83.74
Our	-	**98.10**	**91.96**	**90.95**	**91.45**	**84.25**

1) Experimental Results of LEVIR-CD: The experimental results of various methods on the LEVIR-CD dataset are presented in Table I. As can be seen from Table I, the proposed DF2Net achieves excellent performance. Specifically, DF2Net achieves optimal values for five evaluation metrics (OA, Recall, $F1$, IoU). Compared with EFICNN, which achieved suboptimal values for the four metrics, DF2Net's $F1$ and IoU are higher by 0.30% and 0.51%, respectively. Therefore, it can be concluded that the overall performance of DF2Net is better than that of other mainstream methods, highlighting its potential in remote sensing image change detection tasks.

Fig. 5 shows examples of the visualization of the change detection results of the proposed DF2Net with other comparative methods on the LEVIR-CD dataset, where (1)–(3) represent three sets of samples with building changes at different scales, respectively. In the visualization results, white pixels indicate correctly predicted change pixels, black pixel represents the background pixel, red pixels represent pixels that misclassify non-changes as changes, and blue pixels represent pixels that misclassify changes as non-changes. In Samples 1 and 2, it can be seen that methods such as FC-Siam-diff, ICIF-Net, and CDNeXt are deficient in predicting the completeness of small-scale change regions, compared to DF2Net, which predicts the building change features more accurately, with significantly fewer red and blue pixels. In Sample 3, facing the large-scale building change scene, many methods (such as SNUNet, ICIF-Net, AERNet, etc.) perform poorly in detail restoration, and there is a large deviation of the change region in the prediction results. In contrast, DF2Net is able to effectively differentiate the change region from the background, producing more accurate change detection results.

Therefore, the visualization results further validate the effectiveness of DF2Net. In comparison to other methods, DF2Net demonstrates outstanding performance across building change scenarios at varying scales, with a marked reduction in the number of red and blue pixels in its predictions. This performance can be attributed to the collaborative design of three modules, in which the DFAM module enables the model to capture the building change features more accurately, and at the same time to obtain richer multiscale change information through the global–local feature fusion mechanism. Additionally, the use of pseudolabels establishes a target orientation for feature learning, further enhancing the model's robustness and reliability.

2) Experimental Results of WHU-CD: The experimental results of various methods on the WHU-CD dataset are shown in Table II. Experimental results demonstrate that the proposed DF2Net performs in this article performs well on multiple indicators, achieving the optimal values of OA (97.53%), recall (91.25%), $F1$ (91.92%), and IoU (85.04%). Compared with EFICNN, which achieved suboptimal values for three indicators, OA increased by 0.12%, $F1$ increased by 0.35%, and IoU increased by 0.59%. Compared with other methods, DF2Net achieves a good balance between precision and recall, with a precision of 92.59% and a recall of 91.25%. Although SEIFNet achieves the highest precision (95.56%), its recall is only 86.06%, indicating that this method has obvious missed judgments. In contrast, DF2Net can better balance the problems of misjudgment and missed judgment, showing more balanced performance. Overall, the experimental results in Table II fully verify the excellent performance of DF2Net on the WHU-CD dataset, further validating its excellent potential and practical value in remote sensing image change detection tasks.

Fig. 6 presents the visualization of the change detection results for different methods on the WHU-CD dataset. As can be seen from Fig. 6, compared with other methods, DF2Net more accurately identifies the change regions and reduces both false detections and missed detections, demonstrating significant advantages across various scenarios. Specifically, in the small- and medium-scale change scenarios of Samples 1 and 2, methods such as FC-Siam-diff, AMTNet, and CDNeXt exhibit deficiencies in delineating the boundary details of the change regions, resulting in more frequent misjudgments and undetected changes. In contrast, DF2Net significantly reduces the number of red and blue pixels, and the predicted change regions are highly consistent with the ground truth. This indicates that DF2Net can predict more accurate change detection

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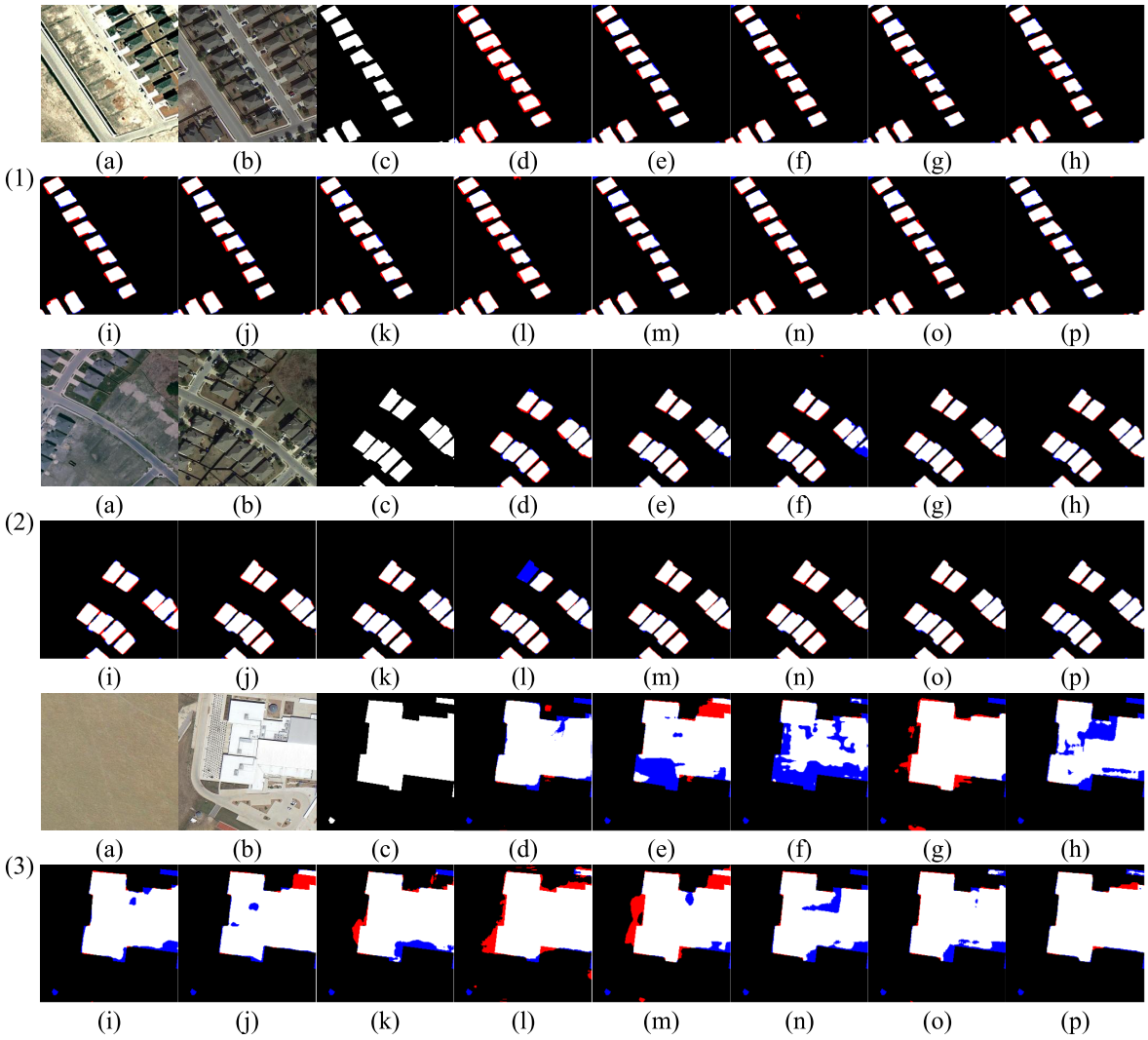


Fig. 5. (1)–(3) Visual results of different models on LEVIR-CD. (a) Image1. (b) Image2. (c) Label. (d) FC-Siam-diff. (e) SNUNet. (f) ICIF-Net. (g) LSNet. (h) AERNet. (i) AMTNet. (j) GeSANet. (k) TinyCD. (l) CDNeXt. (m) SEIFNet. (n) ISDANet. (o) EFICNN. (p) Our (note: White represents TP, black represents TN, red represents FP, and blue represents FN in the figure).

TABLE II
COMPARISON WITH OTHER METHODS ON THE WHU-CD DATASET

Methods	Publication	OA(%)	P(%)	R(%)	F1(%)	IoU(%)
FC-Siam-diff	2018, ICIP	93.58	79.24	79.03	79.14	65.47
SNUNet	2022, GRSL	96.91	91.41	88.24	89.80	81.48
ICIF-Net	2022, TGRS	96.39	92.22	83.67	87.73	78.15
LSNet	2022, IGARSS	97.25	<u>93.41</u>	88.39	90.83	83.21
AERNet	2023, TGRS	<u>97.42</u>	93.13	89.88	91.48	84.29
AMTNet	2023, ISPRS	95.97	88.20	85.26	86.70	76.53
GeSANet	2023, TGRS	96.95	92.16	87.70	89.87	81.61
TinyCD	2023, NCA	97.02	91.89	88.46	90.14	82.06
CDNeXt	2024, JAG	95.15	86.04	81.81	83.87	72.22
SEIFNet	2024, TGRS	97.23	**95.56**	86.06	90.56	82.75
ISDANet	2025, TGRS	97.24	91.64	90.34	90.97	83.44
EFICNN	2025, TGRS	97.41	91.94	<u>91.21</u>	<u>91.57</u>	<u>84.45</u>
Our	-	**97.53**	92.59	**91.25**	**91.92**	**85.04**

when confronted with small- and medium-scale change scenarios. Sample 3 corresponds to a large-scale building change scenario, which imposes higher demands on the model's region perception and detail restoration capabilities.

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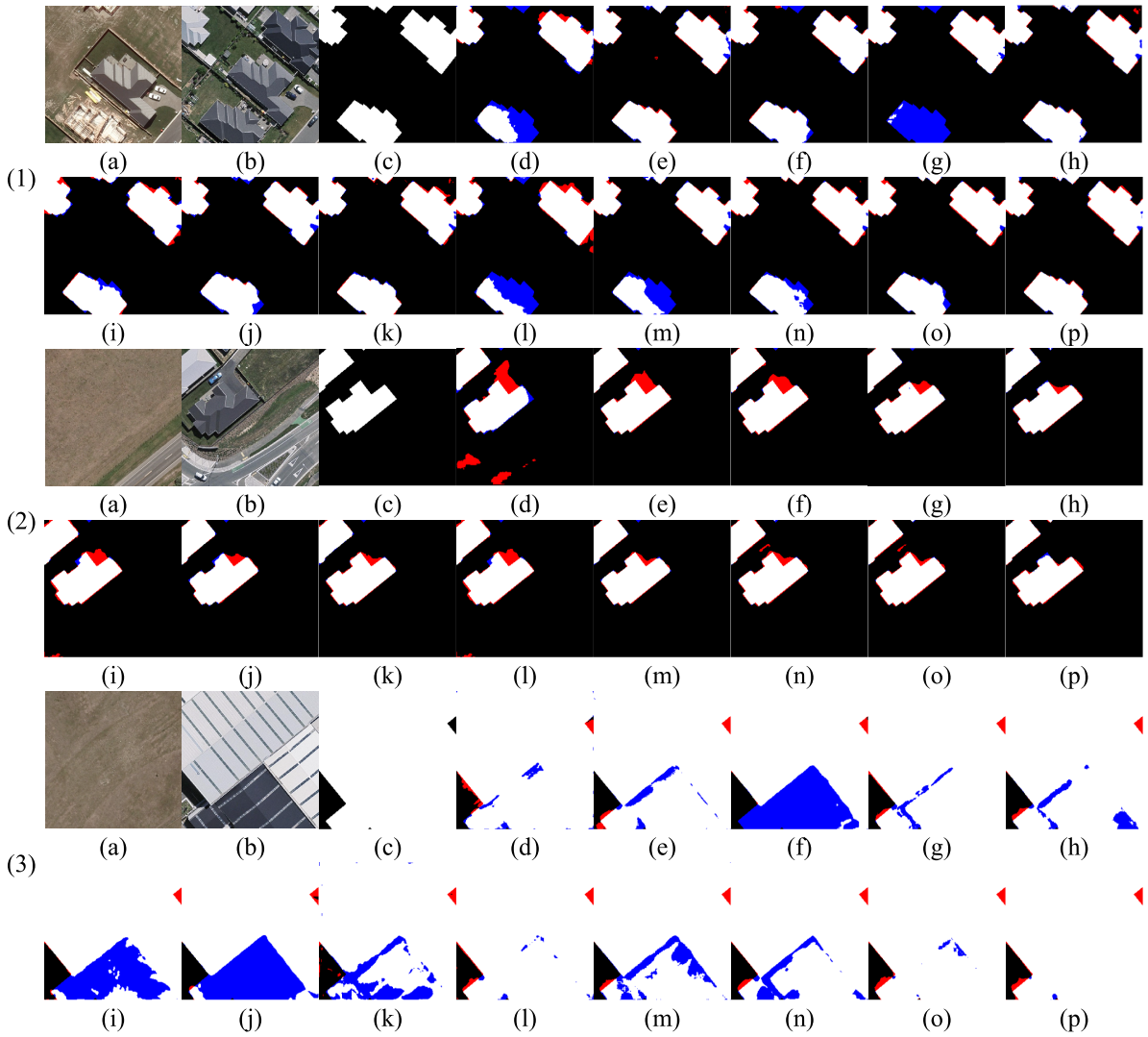


Fig. 6. (1)–(3) Visual results of different models on WHU-CD. (a) Image1. (b) Image2. (c) Label. (d) FC-Siam-diff. (e) SNUNet. (f) ICIF-Net. (g) LSNet. (h) AERNet. (i) AMTNet. (j) GeSANet. (k) TinyCD. (l) CDNeXt. (m) SEIFNet. (n) ISDANet. (o) EFICNN. (p) Our (note: White represents TP, black represents TN, red represents FP, and blue represents FN in the figure).

TABLE III
COMPARISON WITH OTHER METHODS ON THE HRCUS-CD DATASET

Methods	Publication	OA(%)	P(%)	R(%)	F1(%)	IoU(%)
FC-Siam-diff	2018, ICIP	93.87	65.96	54.45	59.65	42.50
SNUNet	2022, GRSL	94.52	70.51	58.76	64.11	47.17
ICIF-Net	2022, TGRS	94.22	66.56	62.04	64.22	47.30
LSNet	2022, IGARSS	94.50	66.77	67.49	67.13	50.52
AERNet	2023, TGRS	94.72	65.09	**78.95**	<u>71.35</u>	<u>55.46</u>
AMTNet	2023, ISPRS	93.98	64.28	62.32	63.28	46.29
GeSANet	2023, TGRS	<u>95.35</u>	**74.10**	67.92	70.88	54.89
TinyCD	2023, NCA	95.08	69.88	<u>71.99</u>	70.92	54.95
CDNeXt	2024, JAG	95.23	72.31	69.18	70.72	54.70
SEIFNet	2024, TGRS	94.86	67.82	72.92	70.28	54.18
ISDANet	2025, TGRS	95.04	70.01	70.75	70.38	54.29
EFICNN	2025, TGRS	95.16	72.08	68.44	70.21	54.10
Our	-	**95.46**	<u>74.04</u>	70.70	**71.96**	**56.20**

Most methods (such as ICIF-Net, GeSANet, and TinyCD) show significant limitations in this scenario, making it difficult to accurately predict the change regions, with numerous omissions and substantial biases in the prediction results.

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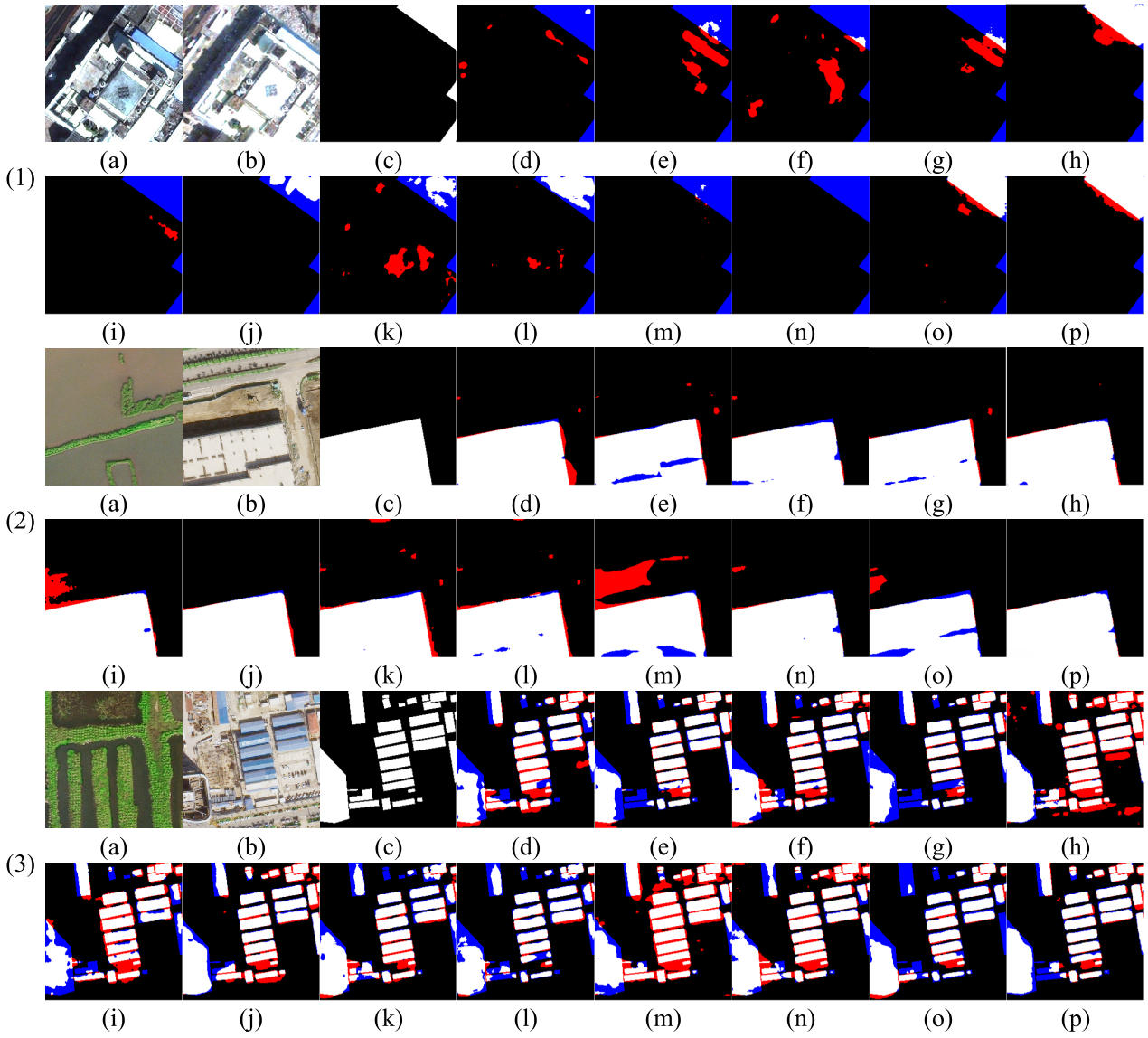


Fig. 7. (1)–(3) Visual results of different models on HRCUS-CD. (a) Image1. (b) Image2. (c) Label. (d) FC-Siam-diff. (e) SNUNet. (f) ICIF-Net. (g) LSNet. (h) AERNet. (i) AMTNet. (j) GeSANet. (k) TinyCD. (l) CDNeXt. (m) SEIFNet. (n) ISDANet. (o) EFICNN. (p) Our (note: White represents TP, black represents TN, red represents FP, and blue represents FN in the figure).

However, DF2Net effectively aggregates multiscale features through the GFDM module, enabling it to still effectively distinguish the change region and background in this scenario. It predicts results with higher integrity, which indicates that the method has good generalization ability when facing large-scale complex scenarios.

3) Experimental Results of HRCUS-CD: Unlike the previous two datasets, the HRCUS-CD dataset covers more complex building change scenarios, and the diversity and complexity of its samples significantly increase the difficulty of change detection, which puts higher requirements on the feature extraction capability, robustness, and generalization ability of the model. The experimental results of various methods on the HRCUS-CD dataset are presented in Table III.

The results indicate that the proposed DF2Net still demonstrates significant advantages on the HRCUS-CD dataset, achieving optimal values in the three key indicators of OA, $F1$, and IoU. Compared to AERNet, which achieves the second-best values for $F1$ and IoU, DF2Net improves $F1$ and IoU by 0.61% and 0.74%, respectively. While DF2Net did not reach the highest values for precision and R\recall, it maintains a good balance. Whereas AERNet achieves the best recall (78.95%), but its precision is only 65.09%, which reflects that this method may have more misjudgments. In contrast, DF2Net effectively balances the problems of misjudgment and omission, performs better in overall performance, and can correctly detect more building change information than other methods.

Fig. 7 presents visual examples of different methods on the HRCUS-CD dataset. As shown in Fig. 7, despite the complex change scenarios in this dataset, DF2Net still provides the best visual results among all methods. Change detection in complex scenes puts forward high requirements for the

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TABLE IV
ABLATION STUDY RESULTS OF DIFFERENT MODULES ON THREE DATASETS

Methods				LEVIR-CD			WHU-CD			HRCUS-CD		
Backbone	DFAM	GFDM	PDM	OA (%)	F1 (%)	IoU (%)	OA (%)	F1 (%)	IoU (%)	OA (%)	F1 (%)	IoU (%)
√	×	×	×	97.76	89.97	81.77	96.82	89.58	81.12	95.04	70.38	54.29
√	√	×	×	97.84	90.27	82.26	97.15	90.75	83.06	95.23	70.89	54.91
√	×	√	×	97.80	90.12	82.02	97.11	90.61	82.83	95.12	70.72	54.70
√	×	×	√	97.77	90.09	81.97	97.07	90.49	82.64	95.20	70.81	54.81
√	√	√	×	97.99	90.91	83.34	97.42	91.37	84.12	95.36	71.36	55.47
√	×	√	√	97.89	90.61	82.83	97.35	91.34	84.06	95.37	71.38	55.49
√	√	×	√	97.92	90.65	82.90	97.43	91.43	84.20	95.40	71.64	55.81
√	√	√	√	**98.10**	**91.45**	**84.25**	**97.53**	**91.92**	**85.04**	**95.46**	**71.96**	**56.20**

model's multiscale perception ability and detail representation. In Samples 1 and 2, most comparison methods (such as FC-Siam-diff, ICIF-Net, AMTNet, etc.) exhibit significant missed judgments. In contrast, DF2Net shows obvious advantages in capturing change integrity and subtle details, with more complete white pixel areas and a smaller number of red and blue pixels in the prediction results, which can more accurately depict the shape and position of the building. In Sample 3, since the region changes from farmland to a factory and there is a lot of redundant information interference in the image, the detection results of all methods would not be ideal. However, compared to other methods, DF2Net still performs well, which further verifies its robustness in multiscale change scenarios. Overall, DF2Net outperforms other existing methods on the HRCUS-CD dataset. This indicates that the method has significant advantages in coping with the complexity and diversity of changes in remote sensing images, and once again demonstrates its great potential in change detection tasks.

IV. DISCUSSION

A. Ablation Experiments

In this article, our proposed DF2Net has three innovative designs: DFAM, GFDM, and PDM. To further verify the impact of each module on model performance, we conducted comprehensive ablation experiments. Among them, when DFAM is not used, the difference features are obtained by direct difference between the twin branches; when GFDM is not used, two adjacent layer features are upsampled and concatenated, and then two 3×3 convolutions are applied to extract features and pass them to the next layer; when PDM is not used, the output of the last decoder layer is directly taken as the final output of the model and the loss is calculated with the label. Table IV shows the ablation experiment results on the LEVIR-CD, WHU-CD, and HRCUS-CD datasets.

It can be seen from the results that when only the backbone is used, the model's performance on various indicators on the three datasets is lower. After the introduction of the DFAM module, the $F1$-score and IoU of all datasets significantly improved, especially on the WHU-CD dataset, where the $F1$-score increased from 89.58% to 90.75%, and IoU improved by 1.94%. This indicates that the DFAM module can effectively extract the key information in the frequency domain and thus enhance the feature representation of change regions. When the GFDM module is introduced alone, the model's performance on all three datasets also improves to varying degrees. This demonstrates that the GFDM module can effectively aggregate global and local features, thus capturing multiscale building change information. When combined with the PDM module alone, various indicators of different datasets can also be significantly improved, strongly validating the effectiveness of this module. Therefore, combining any one of the three modules enhances the model's performance. When any two of the three modules are combined, performance improves further compared to using a single module. Ultimately, when the DFAM, GFDM, and PDM modules are combined at the same time, the DF2Net achieves optimal overall performance on all three datasets, which fully proves the effectiveness of each module and the synergy between them.

B. Comparison With the MFGatt Module

To highlight the advantages of the DFAM module more intuitively, this section compares it with the multi-neighborhood frequency gate attention (MFGatt) [33]. Specifically, the DFAM in DF2Net is replaced with MFGatt and denoted as DF2Net-MFGatt. The two networks are trained and tested on the LEVIR-CD dataset with the same experimental settings. The experimental results show that DF2Net achieves an $F1$ of 91.45% and an IoU of 84.25%, while DF2Net-MFGatt obtains an $F1$ of 91.08% and an IoU of 83.62%, and these results further demonstrate the superiority of DFAM.

Fig. 8 presents the heatmap results of the DFAM module and the MFGatt module. As shown in Fig. 8, the DFAM module demonstrates a superior ability to focus on change regions compared to MFGatt. Specifically, by comparing column (d) (the output of the first DFAM module in DF2Net) and column (f) (the output of the first MFGatt module in DF2Net-MFGatt), it can be observed that during the shallow feature extraction stage, DF2Net with DFAM is more sensitive to building change regions and generates higher heat values

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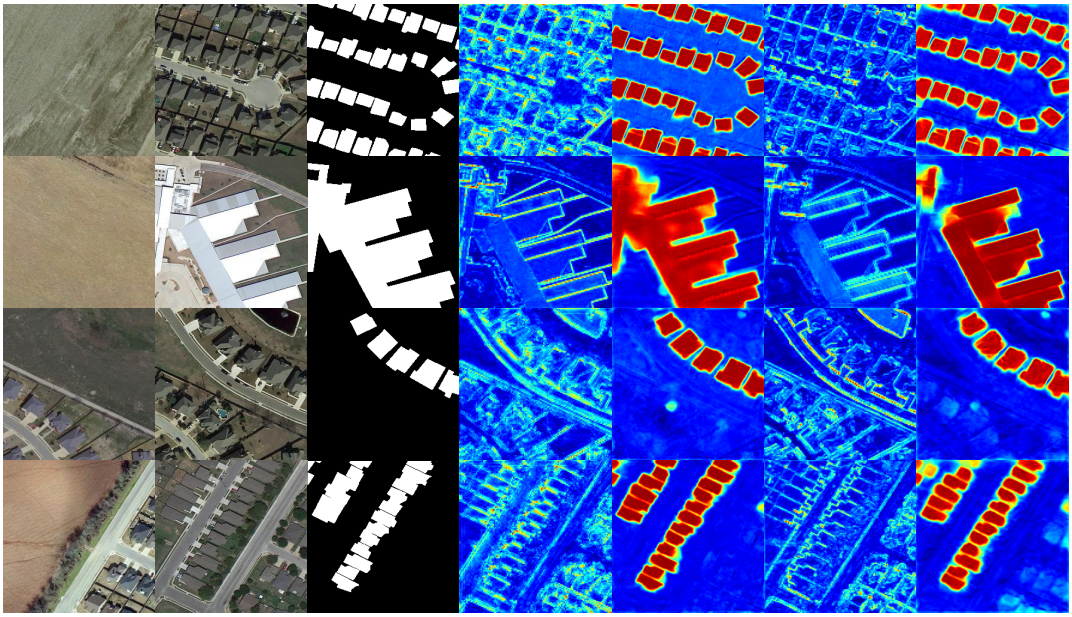


Fig. 8. Heatmap results of the two modules on the LEVIR-CD dataset. (a) Image1. (b) Image2. (c) Label. (d) Output of the first DFAM in DF2Net. (e) Final output of DF2Net. (f) Output of the first MFGatt in DF2Net-MFGatt. (g) Final output of DF2Net-MFGatt.

TABLE V
COMPARISON OF DIFFERENT METHODS OF SUPERVISION

Supervision method	OA(%)	F1(%)	IoU(%)
Real-label	98.03	91.28	83.97
Pseudo-label	**98.10**	**91.45**	**84.25**

TABLE VI
COMPARISON OF DIFFERENT WEIGHT COEFFICIENTS ON THE LEVIR-CD DATASET

Weight Coefficient (α, β)	OA(%)	F1(%)	IoU(%)
(α=0.5, β=0.5)	**98.10**	**91.45**	**84.25**
(α=0.6, β=0.4)	98.06	91.35	84.09
(α=0.7, β=0.3)	98.02	91.26	83.95

in these areas compared to MFGatt. This suggests that DFAM exhibits stronger sensitivity and attention to change regions during the initial feature extraction, thereby more effectively guiding the network to focus on critical change information.

Furthermore, by comparing column (e) (the final output of DF2Net) with column (g) (the final output of DF2Net-MFGatt), the advantage of the DFAM module is even more obvious. The heatmaps generated by DFAM closely match the true change regions indicated in the label, with smoother and clearer boundaries, effectively reducing unnecessary noise and false change information. In contrast, the boundaries of the change areas in the heat map of MFGatt are relatively blurred, and there is unnecessary diffusion of heat values in some areas. This indicates that thanks to the effective utilization of frequency domain information, the DFAM module captures subtle change features more accurately, and thus is able to highlight difference features more effectively, providing more reliable and accurate information for subsequent prediction of change pixels.

C. Superiority of Pseudolabel Supervision

To verify the effectiveness of the proposed pseudolabel supervision in the PDM module, we replace the pseudolabel supervision in the PDM module with the real-label supervision and train and test under the same conditions. The experimental results are shown in Table V. When real labels are used to supervise the middle layer features, the $F1$ and IoU are 91.28% and 83.97%, respectively, whereas when pseudolabels are used for supervision, the $F1$ and IoU are improved to 91.45% and 84.25%, respectively. This result indicates that pseudolabel supervision is more effective in guiding feature learning compared to real-label supervision. Although real label supervision fixes a specific learning objective during feature learning, it may introduce additional noise that affects the stable learning of intermediate layer features, which in turn leads to a significant deviation from the final prediction. In contrast, PDM utilizes the final output of the model as pseudolabels to guide the feature learning of the middle layer, providing progressive supervision signals to the middle layer. This allows the model to more effectively correct potentially erroneous or underfitted features in the middle layer, thus helping to progressively align middle-layer features to the final predicted features.

D. Discussion on the Coefficients of the Loss Function

In Section II-D, we elaborate on the loss function and set the weight coefficients α and β, respectively, for L_{pre} and L_{pse}. In this section, to further explore the impact of different weighted

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TABLE VII
COMPARISON OF MODEL COMPLEXITY OF OTHER METHODS

Method	Parameters (M)	FLOPs (G)	F1 (%)
FC-Siam-diff	1.35	4.28	83.23
SNUNet	12.04	46.81	89.88
ICIF-Net	34.23	24.55	89.04
LSNet	1.11	6.02	89.97
AERNet	25.36	12.82	90.63
AMTNet	24.88	21.39	89.38
GeSANet	36.50	8.69	90.30
TinyCD	**0.29**	**1.54**	89.19
CDNeXt	39.44	18.62	86.94
SEIFNet	28.591	8.32	90.91
ISDANet	6.86	6.66	91.05
EFICNN	21.33	91.44	91.15
Our	8.96	16.18	**91.45**

combinations of predicted loss and pseudolabel supervised loss on DF2Net performance, we tested three typical coefficient combinations on the LEVIR-CD dataset under the condition that the sum of the weight coefficients is 1: ($\alpha = 0.5, \beta = 0.5$), ($\alpha = 0.6, \beta = 0.4$), ($\alpha = 0.7, \beta = 0.3$). The experimental results are shown in Table VI.

Based on the experimental results shown in Table VI, we observed that the coefficient combination of $\alpha = 0.5$ and $\beta = 0.5$ achieved optimal performance (OA = 98.10%, $F1$ = 91.45%, IoU = 84.25%). The superiority of this coefficient combination stems from the complementary roles of L_{pre} and L_{pse} in feature learning. L_{pre} directly supervises the final prediction layer, ensuring that the model output aligns with the ground truth. In contrast, L_{pse} leverages the final output as pseudolabels to supervise intermediate decoder layers, narrowing the gap between intermediate features and the final output. Equal weighting ensures that both parts of the loss are not overlooked: L_{pre} guarantees the accuracy of the final prediction, while L_{pse} optimizes intermediate features, fully leveraging the synergy between modules.

When the weight of L_{pre} increases, model performance declines due to weakened L_{pse} supervision, which leads to suboptimal optimization of intermediate features. For example, when $\alpha = 0.7$ and $\beta = 0.3$, the $F1$ and IoU drop by 0.19% and 0.30%, respectively, indicating that excessive reliance on L_{pre} affects the model's ability to optimize intermediate features. In conclusion, the coefficient combination of $\alpha = 0.5$ and $\beta = 0.5$ achieves the optimal balance between L_{pre} and L_{pse}, and is therefore adopted as the final parameter setting.

E. Model Complexity Analysis

Model complexity is a key indicator in assessing the performance of a model. It reflects the model's demand for computing resources and memory, and mainly includes two aspects: Parameters and FLOPs. In Table VII, we list the model complexity of the proposed DF2Net with other comparative methods. Furthermore, to intuitively evaluate the balance between model complexity and accuracy, we selected the $F1$-score of these methods on the LEVIR-CD dataset as the accuracy metric.

As shown in Table VII, the early change detection models (such as FC-Siam-diff) only use convolution operations and simple feature fusion methods. Although the model have fewer Parameters and lower FLOPs, its accuracy is also lower, with the $F1$-score of only 83.23%, significantly lower than DF2Net. The lightweight networks (such as LSNet and TinyCD) have relatively simple structures and maintain a very low number of parameters, but their performance is also inferior to DF2Net. In contrast, DF2Net has 8.96M parameters and 16.18G FLOPs, which are significantly fewer and lower, respectively, than those of some more complex models (such as SNUNet, ICIF-Net, and EFICNN). This demonstrates that DF2Net maintains a reasonable level of memory and computational resource consumption while providing excellent performance.

In summary, DF2Net not only performs well in accuracy but also has high computational efficiency and achieves a good balance between model complexity and accuracy. This makes DF2Net highly advantageous in practical applications, especially in resource-constrained scenarios, but further improving the training and inference speed while ensuring the performance of the model remains the focus of future work.

V. CONCLUSION

This article proposes a novel DF2Net for building change detection in remote sensing images. Through the multimodule collaborative design, the DF2Net network significantly enhances the detection capability of change regions and the model's generalization performance. We designed the DFAM module, which combines the Fourier transform and self-attention mechanisms to extract key information from differential features in the frequency domain, thereby enhancing the feature expression of change regions. Meanwhile, the GFDM module is constructed to dynamically fuse global and local features, enabling the capture of multiscale change information and enhancing the model's adaptability to complex scenes. Additionally, we introduced the PDM module, which imposes pseudolabel constraints on the middle layers of the decoder, reducing the feature deviation between the middle layers and the final output, thereby further improving the model's convergence speed and detection accuracy. A series of experimental results on three mainstream change detection datasets (LEVIR-CD, WHU-CD, and HRCUS-CD) show that DF2Net outperforms other methods and significantly improves the accuracy of change detection and the adaptability to complex scenes. Nevertheless, we recognize that there is still room for improvement. Future work will focus on reducing model complexity, improving inference speed, and exploring multimodal heterogeneous change detection methods to further advance research and applications in this field.

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Like Zhao received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2018.

She is currently an Associate Professor with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou, Henan, China. Her research interests include photogrammetry, intelligent remote sensing, and image interpretation.



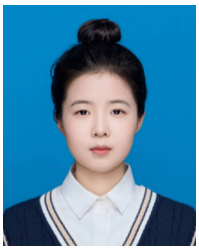
Mingwang Yang received the master's degree in computer technology from Henan University of Technology, Zhengzhou, China, in 2025.

His research interests include intelligent interpretation of remote sensing images and computer vision (CV).



Yiran Zhao is currently pursuing the bachelor's degree in the Internet of Things Engineering with Henan University of Technology, Zhengzhou, China.

Her research interests include intelligent interpretation of remote sensing images.



Yuqing Liu received the bachelor's degree in the Internet of Things Engineering from Xinyang Agriculture and Forestry University, Xinyang, China, in 2024. She is currently pursuing the degree with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou, Henan, China.

Her research interests include intelligent interpretation of remote sensing images.



Huawei Jiang was born in Jiyuan, Henan, in 1970. He is currently a Professor and Doctoral Supervisor with the School of Information Science and Engineering, Henan University of Technology. His main research interests include optimization computing, intelligent information processing, and machine learning.



Zhen Yang received the Ph.D. degree in geographical information system from China University of Mining and Technology, Beijing, China, in 2019.

He is currently a Lecturer with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou, China. His research interests include digital image processing and remote sensing of resources and the environment.

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